

AIRLINE FLIGHT & PASSENGER ANALYSIS REPORT

1. Executive Summary

This project analyzes airline flight and passenger data to understand airline performance, passenger volume, revenue, flight status, cancellations, booking channels, travel classes, routes, aircraft usage, delays, and passenger satisfaction. The analysis follows an end-to-end data analytics workflow using Excel for initial cleaning, Python for reproducible data preparation and exploratory analysis, SQL for structured querying, and Power BI for interactive visualization.

After data cleaning and validation, the final dataset contained 34,935 records and 25 columns. The analysis identified 34,930 unique passengers and 8,825 unique flights, with total recorded ticket revenue of ₹62.88 million. The average ticket fare was approximately ₹1,800 and average delay was 5.45 minutes. The Power BI dashboard highlights the strongest business-level patterns and provides a concise view of airline, route, channel, status, and monthly performance.

2. Project Objectives

- Measure overall passenger, flight, revenue, fare, and delay performance.
- Compare airlines on passenger volume, revenue, flight activity, and cancellation rate.
- Understand flight status and operational performance.
- Identify high-volume and high-delay routes.
- Analyze booking channel and travel-class patterns.
- Examine monthly passenger and revenue trends.
- Evaluate aircraft-type performance.
- Study the relationship between passenger satisfaction and delays.
- Build an interactive Power BI dashboard suitable for portfolio presentation.

3. Dataset Overview

Item	Details
Final records	34,935
Columns	25
Unique passengers	34,930
Unique flights	8,825
Main domains	Passengers, flights, routes, fares, delays, cancellations, satisfaction
Final analytical tools	Excel, Python, SQL, Power BI

4. Data Preparation & Cleaning

The raw workbook intentionally contained realistic data-quality issues so that the project could demonstrate an end-to-end cleaning workflow. Cleaning was performed in stages rather than relying on a single tool.

4.1 Excel Cleaning

- Removed 62 duplicate records during the initial Excel cleaning stage.
- Filled blank Booking_Channel values with 'Unknown'.

- Inspected missing values, inconsistent formatting, and currency representations.
- Saved the cleaned workbook for downstream Python analysis.

4.2 Python Cleaning

- Loaded the Excel-cleaned dataset with Pandas.
- Audited missing values across all 25 columns.
- Handled missing Passenger_Age, Ticket_Fare, Baggage_KG, and Passenger_Satisfaction using median imputation.
- Standardized Ticket_Fare by removing currency symbols/separators and converting it to numeric format.
- Cleaned Delay_Minutes values containing text such as 'min'.
- Converted analytical fields to appropriate numeric and date/time data types.
- Validated duplicate records and removed 3 remaining exact duplicates.
- Standardized categorical values such as airline names and travel classes.
- Final validation confirmed zero missing values and 34,935 final records.

5. Analytical Methodology

1. Descriptive analysis was used to establish overall KPIs and dataset-level patterns.
2. Group-by and aggregation analysis was used to compare airlines, routes, channels, classes, and aircraft types.
3. SQL was used to reproduce and validate important business questions using MySQL.
4. Power BI measures and visuals were used to create an interactive management-style dashboard.
5. Results were cross-checked across Python, SQL, and Power BI wherever practical.

6. Key Performance Indicators

KPI	Result
Unique Passengers	34,930
Unique Flights	8,825
Total Revenue	₹62,883,106
Average Ticket Fare	₹1,800
Average Delay	5.45 minutes
On-Time Rate	97.58%
Flight Records	34,935

7. Airline Performance Analysis

Airline	Passengers	Flights	Revenue	Avg Delay
IndiGo	11,851	6,596	₹21,331,906	5.40 min
Air India	7,069	4,911	₹12,729,600	5.49 min
Vistara	5,638	4,153	₹10,150,200	5.37 min
SpiceJet	4,449	3,496	₹8,008,200	5.38 min
Go First	3,176	2,683	₹5,718,600	5.44 min
Akasa Air	2,747	2,394	₹4,944,600	5.83 min

- IndiGo recorded the highest passenger volume and total revenue in the dataset.
- Akasa Air had the highest average delay at approximately 5.83 minutes, while Vistara had the lowest at approximately 5.37 minutes.
- Average delay differences between airlines were relatively small, so airline-level delay should not be treated as the strongest differentiator in this dataset.
- Revenue per flight was highest for IndiGo at approximately ₹3,234 per flight.

8. Flight Status & Cancellation Analysis

Flight Status	Records
On Time	28,240
Delayed	5,182
Cancelled	1,513

The dashboard shows a strong on-time operating profile, with an on-time rate of 97.58% based on distinct flights. Cancellation rate varied by airline, with IndiGo at approximately 7.55% and Akasa Air at approximately 4.80% in the flight-level analysis.

Airline	Total Flights	Cancelled Flights	Cancellation Rate
IndiGo	6,596	498	7.55%
Vistara	4,153	248	5.97%
Air India	4,911	288	5.86%
SpiceJet	3,496	193	5.52%
Go First	2,683	144	5.37%
Akasa Air	2,394	115	4.80%

9. Passenger & Commercial Analysis

9.1 Travel Class

Travel Class	Passengers	Revenue	Avg Fare
Economy	26,450	₹47,615,400	₹1,800
Business	7,160	₹12,891,600	₹1,800
First	1,320	₹2,376,106	₹1,800

- Economy represented the largest passenger segment and generated the highest total revenue.
- The similar average fare across classes suggests that the dataset is more useful for volume and mix analysis than for comparing class-level pricing differences.

9.2 Booking Channel

Booking Channel	Passengers	Revenue
Website	11,757	₹21,166,306
Mobile App	10,502	₹18,905,400
Travel Agent	5,847	₹10,526,400
Airport Counter	3,428	₹6,170,400
Corporate	3,187	₹5,736,600
Unknown	210	₹378,000

- Website was the largest booking channel, followed closely by the Mobile App.
- Digital channels together represent the dominant booking pattern in the dataset.

10. Route Analysis

Route	Passengers	Flights	Revenue
BOM → JAI	439	426	₹790,200
GOI → CCU	433	429	₹779,400
HYD → CCU	432	423	₹777,600
GOI → HYD	422	411	₹759,600
PNQ → CCU	421	410	₹759,600
BOM → CCU	421	411	₹757,800
PNQ → HYD	419	411	₹754,200

JAI → GOI	413	408	₹743,400
HYD → MAA	413	400	₹743,400
HYD → BLR	411	399	₹739,800

The top passenger-volume routes were relatively close in size, with BOM → JAI ranking first. For delay analysis, only routes with at least 50 distinct flights were considered to avoid overemphasizing routes with very few observations.

10.1 Most Delayed Frequent Routes

Route	Flights	Avg Delay
BOM → GOI	347	7.47 min
DEL → PNQ	364	6.66 min
GOI → AMD	393	6.60 min
CCU → HYD	367	6.59 min
HYD → BLR	399	6.53 min

11. Monthly Trend Analysis

Month	Passengers	Revenue
January	3,029	₹5,452,306
December	3,021	₹5,437,800
March	3,016	₹5,428,800
May	2,980	₹5,364,000
August	2,964	₹5,337,000
October	2,953	₹5,317,200
April	2,875	₹5,175,000
July	2,866	₹5,158,800
September	2,849	₹5,128,200
November	2,837	₹5,110,200
June	2,825	₹5,086,800
February	2,715	₹4,887,000

- January recorded the highest passenger volume and revenue among the months analyzed.
- February recorded the lowest passenger volume and revenue.
- The monthly pattern is relatively stable rather than showing extreme seasonality.

12. Aircraft Type Analysis

Aircraft Type	Passengers	Flights	Revenue	Avg Delay
Airbus A320	11,893	6,588	₹21,411,106	5.56 min
Boeing 737	10,647	6,244	₹19,170,000	5.49 min
Airbus A321	6,135	4,442	₹11,043,000	5.27 min
Boeing 787	3,498	2,910	₹6,296,400	5.31 min
Airbus A350	2,757	2,352	₹4,962,600	5.35 min

- Airbus A320 had the highest passenger volume and total revenue.
- Airbus A321 had the lowest average delay among the aircraft types shown.

13. Passenger Satisfaction & Delay

Satisfaction Score	Passengers	Average Delay
1	860	0.00 min
2	636	2.60 min
3	4,330	8.43 min
4	17,516	5.93 min
5	11,593	4.17 min

The relationship between satisfaction and delay is not strictly linear in this dataset. Satisfaction score 3 shows the highest average delay at 8.43 minutes, while the highest satisfaction group (5) has an average delay of 4.17 minutes. Because the pattern is not monotonic, satisfaction should be interpreted alongside other passenger and operational factors.

14. Power BI Dashboard

The final Power BI dashboard was designed as a concise management-style overview rather than a collection of excessive visuals. It includes six KPI cards and the most relevant analytical views.

- Total Passengers
- Total Flights
- Total Revenue
- Average Ticket Fare
- Average Delay
- On-Time Rate
- Passengers by Airline
- Revenue by Airline
- Flight Status
- Cancellation Rate by Airline
- Passengers by Booking Channel
- Monthly Passenger Trend
- Top 10 Routes by Passengers

15. Business Recommendations

- Monitor cancellation patterns at airline and route level, especially where cancellation rates are relatively high.
- Prioritize operational review of frequently operated routes with above-average delays, such as BOM → GOI.
- Continue strengthening website and mobile booking journeys because digital channels account for the largest booking volumes.
- Use route-level passenger and revenue trends to support capacity planning and scheduling decisions.
- Track passenger satisfaction together with delay, cancellation, travel class, and route information rather than using delay alone as a satisfaction predictor.
- Use aircraft-level operational metrics to identify aircraft types and fleets associated with stronger punctuality or higher utilization.

16. Conclusion

This project demonstrates a complete data analytics workflow from raw data preparation to business insights and interactive reporting. The analysis combines Excel, Python, MySQL, and Power BI to evaluate airline operations, passenger behavior, commercial performance, routes, delays, and cancellations. The final dashboard converts the analysis into a compact decision-support view, while the supporting Python and SQL work provides reproducible analytical depth.